

PROPOSAL FOR DISCUSSION

Sadak Setu Digital Stack

Faster DPRs, sound cost checks and time-bound approvals for Jharkhand's roads

Prepared for: **Road Construction Department (Path Nirman Vibhag), Government of Jharkhand**

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₹88 lakh – ₹1.21 cr

complete platform, one-time (all 7 modules)

₹21–32 lakh / yr

hosting, AI usage and maintenance

10 months

to full rollout, in two phases

THE SEVEN MODULES

01	Smart Cost Estimation · Benchmarking against past sanctions <i>Every estimate compared with similar past projects before it reaches the approving officer.</i>	₹15–20 lakh
02	Auto-DPR Drafting · Assisted report generation <i>A complete, standard-format DPR draft in days, from survey data.</i>	₹20–26 lakh
03	Compliance Checker · Automated standards verification <i>Every DPR checked against the rules before it leaves the division.</i>	₹9–12 lakh
04	Cost Assurance · Independent cost verification <i>Three independent checks that catch what a single threshold cannot.</i>	₹5–6 lakh
05	Digital Approval Tracking · DPR workflow with deadlines <i>Every DPR tracked from submission to sanction, with automatic escalation.</i>	₹12–16 lakh
06	Project Knowledge Base · Searchable archive of past projects <i>Twenty-five years of project memory, searchable in seconds.</i>	₹13–19 lakh
07	GIS Link-up · DPR pipeline on RCD's existing map <i>Every proposal and sanction placed on the road map RCD already uses.</i>	₹6–8 lakh

Module prices include each module's share of the shared platform. All figures exclude GST and are indicative until scoped jointly with the Department.

At a glance

Jharkhand's Road Construction Department looks after a 12,736 km network, including about 9,450 km of State Highways, Major District Roads and other state roads, and has ₹6,601 crore for roads in 2026–27. Every project starts with a Detailed Project Report (DPR). Today a DPR is drafted in varying formats, priced against rates spread across the last published Schedule of Rates and later circulars, checked by hand at several levels, and moved as a paper file. Sadak Setu is one platform with seven modules that make each step faster, consistent and auditable, without replacing the engineer's judgement.

#	Module	What it solves	Status today	Phase	One-time
01	Smart Cost Estimation	No benchmark; rates applied by hand from scattered sources	Not started	Phase 1	₹15–20 lakh
02	Auto-DPR Drafting	DPRs rewritten from scratch in varying formats	Not started	Phase 2	₹20–26 lakh
03	Compliance Checker	Checkable errors caught late, at CDO	Manual review only	Phase 2	₹9–12 lakh
04	Cost Assurance	No independent check of estimate reasonableness	Not started	Phase 2	₹5–6 lakh
05	Digital Approval Tracking	Paper file movement; no stage deadlines	File-based	Phase 1	₹12–16 lakh
06	Project Knowledge Base	25 years of DPRs unsearchable in record rooms	Paper archives	Phase 1	₹13–19 lakh
07	GIS Link-up	DPR pipeline not visible on RCD's GIS	GIS portal exists	Phase 2	₹6–8 lakh
Platform-wide items: security audit, AI usage during build, on-site presence in Ranchi and workshops					₹8–14 lakh
Complete platform, one-time					₹88 lakh – ₹1.21 cr
Recurring per year: hosting, AI usage, maintenance for all seven modules					₹21–32 lakh

For scale: the complete platform costs about 0.013–0.018% of RCD's 2026–27 roads allocation. Improving estimate accuracy on a single ₹60 crore project by 2% would recover its full cost.

→ HOW THE SEVEN FIT TOGETHER

One DPR, one digital thread. The Knowledge Base (06) turns past DPRs into data. Smart Cost Estimation (01) benchmarks each new proposal. Auto-DPR Drafting (02) produces a standard draft. The Compliance Checker (03) and Cost Assurance (04) attach evidence before the file moves. Digital Approval Tracking (05) moves it to sanction on a clock. The GIS Link-up (07) shows it all on RCD's existing map. All seven share one login, one workflow engine, one document store and one dashboard, which is why the whole platform costs about as much as a single stand-alone system.

RECOMMENDED START

- **Proof of concept, 6–8 weeks, one division:** Knowledge Base + cost benchmark preview on RCD's own past DPRs. ₹6–8 lakh, adjusted in full against Phase 1 if the Department proceeds.
- **Phase 1, months 1–4 (₹44–62 lakh):** Modules 06, 01 and 05: the data foundation, benchmarking and time-bound approvals.
- **Phase 2, months 5–10 (₹44–59 lakh):** Modules 02, 03, 04 and 07: drafting, automated checks, cost assurance and the map link-up.

How the price is built

The price is built bottom-up from the work involved: about 70 person-months from one team of eight over ten months, at a blended ₹1.1–1.4 lakh per person-month, plus a small set of pass-through costs. There is one platform, one team, one hosting environment, one security audit and one maintenance contract, not seven separate systems.

A EFFORT BY WORKSTREAM

Workstream	Person-months	Low	High
Shared platform: login & roles, audit trail, document store, workflow engine, dashboards, Hindi/English	10	₹11 lakh	₹14 lakh
01 Smart Cost Estimation (incl. digital Schedule of Rates library)	10	₹11 lakh	₹14 lakh
02 Auto-DPR Drafting	13	₹14 lakh	₹18 lakh
03 Compliance Checker	6	₹7 lakh	₹8 lakh
04 Cost Assurance	3	₹3 lakh	₹4 lakh
05 Digital Approval Tracking	8	₹9 lakh	₹11 lakh
06 Project Knowledge Base	6	₹7 lakh	₹8 lakh
07 GIS Link-up (with the existing RCD GIS portal)	4	₹4 lakh	₹6 lakh
Project management, testing, security hardening, user acceptance, training	10	₹11 lakh	₹14 lakh
Team effort	70	₹77 lakh	₹98 lakh
Digitisation of up to ~800 past DPRs (scanning, OCR, quality check)	—	₹4 lakh	₹7 lakh
Independent security audit by a CERT-In-empanelled auditor	—	₹3 lakh	₹5 lakh
AI model usage and cloud test environments during the build	—	₹2 lakh	₹4 lakh
On-site presence in Ranchi, workshops and travel	—	₹3 lakh	₹5 lakh
Complete platform, one-time (module prices are rounded shares of this)		₹88 lakh	₹1.21 cr

B THE DELIVERY TEAM

Role	People	Months	Person-months
Solution lead & project manager (civil engineering + IT)	1	10	10
Highway domain engineer (IRC codes, SoR, DPR rules)	1	8	8
Backend developers	2	10	20
Frontend / mobile developer	1.5	9	13
AI & data engineer	1	8	8
GIS developer	0.5	4	2
QA / test engineer	1	6	6
DevOps & security (part-time)	0.3	10	3
Total			70

C WHY ONE PLATFORM COSTS SO MUCH LESS THAN SEVEN SYSTEMS

- **Built once, used seven times:** login, roles, audit trail, workflow, documents, dashboards and Hindi/English support are shared by every module.
- **One team, not seven:** the same eight people carry the design from Module 06 through to Module 02, with no hand-offs.
- **Reuse what Jharkhand already has:** RCD's GIS portal, the State Data Centre run by JAP-IT and the state's e-Office are integrated, not rebuilt.
- **Open-source stack, no licence fees:** AI is pay-per-use or runs on open models hosted in India, so there are no per-seat charges.
- **One audit, one hosting environment, one maintenance contract:** running costs are 15–18% of the build a year, in line with common government practice.

D PHASES, PAYMENTS AND RUNNING COSTS

Stage	Scope	Duration	Price
Proof of concept	One division; Knowledge Base and benchmark preview on RCD data	6–8 weeks	₹6–8 lakh*
Phase 1	Modules 06, 01, 05 live in all circles	Months 1–4	₹44–62 lakh
Phase 2	Modules 02, 03, 04, 07; three months of hypercare	Months 5–10	₹44–59 lakh
Annual maintenance & hosting	Hosting; AI usage (report drafting, smart search), SMS and e-mail alerts; Annual maintenance	Per year	₹21–32 lakh

* Adjusted in full against Phase 1. Payments are linked to milestones: requirements sign-off 20%, user acceptance 30%, go-live 40%, and 10% after 90 days of stable operation.

IF THE BUDGET IS TIGHTER

Essential package: Modules 06, 01, 03 and 05 (benchmarking, automated checks and time-bound approvals on a searchable archive) for ₹57–81 lakh one-time. Drafting, cost assurance and the GIS link-up can follow when funds allow, with no rework.

Included

- Source code and data owned by the state; source code in escrow
- Training for all roles (up to 10 workshops) and user manuals in Hindi and English
- Three months of hypercare after go-live
- Independent security audit before go-live
- Digitisation of up to ~800 past DPRs

Not included

- GST
- Server hardware at the State Data Centre, if the state hosts it
- Field surveys, traffic counts or drone data collection
- Scanning beyond ~800 DPRs (₹1.5–3 per page)
- Changes in scope after requirements sign-off

Smart Cost Estimation

Benchmarking against past sanctions

Every estimate compared with similar past projects before it reaches the approving officer.

OWNER & PARTNERS RCD: Central Design Organisation (CDO), Chief Engineer (Communication), divisions; Finance	STATUS TODAY Not started. Estimates are built division by division. The last fully published RCD Schedule of Rates appears to be 2021–22, with current rates issued through tender notices, so rate application is manual and uneven.
WHEN Phase 1 (months 1–4)	ONE-TIME PRICE ₹15–20 lakh · running costs shared across the platform

1 THE PROBLEM

- RCD's roads budget for 2026–27 is ₹6,601 crore, yet each estimate is prepared without a systematic comparison to what similar roads actually cost.
- Similar roads in similar terrain can carry very different per-km estimates, with nothing to flag or explain the gap.
- Current rates are spread across the last published SoR, circulars and tender notices, so the same item can be priced differently in two divisions.

2 WHAT WE WILL BUILD

A benchmarking engine trained on RCD's past DPRs and sanction orders. Enter the basics (road class, length, width, terrain, traffic, district) and it returns a benchmark cost and the closest comparable projects.

- **Digital rate library:** Current SoR items, circulars and approved rate analyses in one versioned library, applied automatically.
- **Explainable benchmark:** Shows which items make an estimate higher or lower than its peers, in plain language.
- **Outlier alerts:** Estimates far from the benchmark are flagged with the reason before approval.
- **Review dashboard:** All estimates across circles with comparison charts and a weekly Hindi/English review pack.

WHY THIS IS DIFFERENT

It learns from what RCD actually sanctioned, not from generic averages, and explains every number. The same engine powers Module 04, so cost assurance comes almost free.

3 HOW WE WILL DO IT

Step	Duration	What we do	Output
Discover	3 weeks	Collect past DPRs and sanction orders; agree data access with CDO and Finance.	Data baseline
Pilot	8 weeks	Build rate library and benchmark for the top 3 road types; test in Ranchi circle.	Live comparison dashboard
Roll out	6 weeks	All circles; weekly review pack; connect to Modules 04 and 06.	State-wide benchmarking

4 BENEFITS

For the Department

- Every estimate checked against evidence before sanction.
- If better benchmarking trims even 0.5% off inflated estimates, that is about ₹30 crore a year on a ₹6,000 crore capital outlay.
- Audit trail ready for AG and vigilance review.

For consultants, contractors & citizens

- Clear, consistent rate basis for consultants and contractors.
- Fewer disputes over estimate reasonableness.
- Engineers spend less time looking up rates.

5 PRICE AND KEY RISKS

One-time price: ₹15–20 lakh (includes its share of the shared platform). Hosting, AI usage and maintenance are priced once for the whole platform.

Old DPR data is patchy	Start with the 200 largest recent DPRs; show confidence levels; accuracy rises as data grows.
Seen as policing engineers	Framed as decision support: it strengthens an estimate, it does not reject it.

Auto-DPR Drafting

Assisted report generation

A complete, standard-format DPR draft in days, from survey data.

OWNER & PARTNERS RCD: CDO (design & estimation), divisions, empanelled DPR consultants	STATUS TODAY Not started. DPRs are prepared by divisions or empanelled consultants in varying formats. Pavement design and quantities live in individual spreadsheets.
WHEN Phase 2 (months 5–10)	ONE-TIME PRICE ₹20–26 lakh · running costs shared across the platform

1 THE PROBLEM

- Much of every DPR (design basis, specifications, quantity build-up, standard chapters) is rewritten from scratch each time.
- Pavement design is done in personal spreadsheets; one wrong input carries through to the whole estimate. Mining-traffic corridors make this riskier.
- Forest-land, land-acquisition and utility-shifting sections are often thin, which later causes the delays Jharkhand road projects are known for.

2 WHAT WE WILL BUILD

From survey inputs (alignment, soil CBR, traffic counts, drainage), the system produces a full draft DPR in RCD's standard format, exportable to Word for review and signature.

- **Pavement design engine:** Layer thicknesses computed to IRC:37, including heavy-axle mining traffic; every input and result shown.
- **Bill of Quantities:** Quantities linked to the Module 01 rate library with automatic rate application.
- **Drafted narrative:** Standard chapters and project description drafted by AI, then edited and approved by the engineer.
- **Clearance chapter:** Structured sections for forest land, land acquisition and utilities, so gaps are visible at draft stage.

WHY THIS IS DIFFERENT

Engineering first, AI second: the design and costs come from IRC rules and RCD rates, while AI only drafts the prose. Consultants and divisions submit in one format, which makes review much faster.

3 HOW WE WILL DO IT

Step	Duration	What we do	Output
Baseline	3 weeks	Pick the 3 most common project types; agree the standard format with CDO.	Template library
Pilot	10 weeks	Build design engine, BoQ and drafting; produce 10 DPRs in parallel with the normal process.	10 assisted DPRs
Roll out	8 weeks	All circles and consultants; connect to Modules 03 and 04; training.	Standard DPR pipeline

4 BENEFITS

For the Department

- Drafting time for common project types cut from weeks to days.
- One format across all 24 districts and all consultants.
- Fewer return-for-correction rounds.

For consultants, contractors & citizens

- Consultants get a clear template and instant calculations.
- Projects reach tender sooner.
- Standardised estimates reduce tendering disputes.

5 PRICE AND KEY RISKS

One-time price: ₹20–26 lakh (includes its share of the shared platform). Hosting, AI usage and maintenance are priced once for the whole platform.

Engineers distrust AI text	AI drafts only the narrative; design and costs are rule-based; the engineer signs.
IRC editions change	Design rules are versioned; new editions coded within weeks of notification.

Compliance Checker

Automated standards verification

Every DPR checked against the rules before it leaves the division.

OWNER & PARTNERS RCD: CDO, Quality Control wing, divisions	STATUS TODAY Manual review only. Each level (division, circle, CDO) checks by hand. The same basic errors are caught late, after review time has been spent.
WHEN Phase 2 (months 5–10)	ONE-TIME PRICE ₹9–12 lakh · running costs shared across the platform

1 THE PROBLEM

- Common return reasons (pavement thickness, carriageway width for the road class, drainage sizing, missing chapters) are mechanically checkable, yet checked by hand.
- There is no shared digital checklist, so reviewers apply their own interpretation.
- Forest-land and clearance gaps are spotted late, when they are most expensive to fix.

2 WHAT WE WILL BUILD

A rules engine that tests each DPR against IRC standards and RCD norms, and cites the clause for every failure.

- **Engineering rules:** Pavement thickness for traffic, width by road class, camber, drainage and culvert sizing.
- **Completeness rules:** All required chapters, drawings and annexures present before the file can move.
- **Clearance flags:** Alignment through forest land or sensitive areas flagged early for the clearance process.
- **Review bands:** Warn/fail bands rather than bare pass/fail; overrides need a recorded reason.

WHY THIS IS DIFFERENT

Not a paper checklist: each rule runs automatically, cites the standard and clause, and says exactly what failed and why.

3 HOW WE WILL DO IT

Step	Duration	What we do	Output
Baseline	2 weeks	List the 15 most frequent return reasons with CDO.	Rule catalogue
Pilot	6 weeks	Automate 10 rules; run in shadow mode beside manual review.	Validated rules
Roll out	6 weeks	30+ rules; connect to Modules 02 and 05.	State-wide checker

4 BENEFITS

For the Department

- Checkable errors caught at source, not at CDO.
- Reviewers focus on engineering judgement.
- Every decision documented.

For consultants, contractors & citizens

- Instant feedback to consultants on gaps.
- Predictable review timelines.

5 PRICE AND KEY RISKS

One-time price: ₹9–12 lakh (includes its share of the shared platform). Hosting, AI usage and maintenance are priced once for the whole platform.

Rules too rigid for site conditions	Review bands and documented overrides.
Officers feel second-guessed	Shadow mode for one cycle before it becomes mandatory.

Cost Assurance

Independent cost verification

Three independent checks that catch what a single threshold cannot.

OWNER & PARTNERS RCD: CDO, Finance, Vigilance	STATUS TODAY Not started. An estimate is checked mainly against its own build-up. There is no independent benchmark or peer comparison.
WHEN Phase 2 (months 5–10)	ONE-TIME PRICE ₹5–6 lakh · running costs shared across the platform

1 THE PROBLEM

- A fixed-percentage check cannot tell difficult terrain from an inflated estimate.
- No one compares the item-wise total with an independent benchmark or with per-km costs of similar sanctioned roads.
- Manual changes to system-generated figures are not tracked.

2 WHAT WE WILL BUILD

A verification layer on top of Module 01 that attaches a one-page assurance report to every DPR.

- **Check 1: build-up vs benchmark:** Item-wise total compared with the Module 01 benchmark.
- **Check 2: override tracking:** Any manual change to computed figures is logged and graded.
- **Check 3: peer comparison:** Per-km cost against similar sanctioned roads in the same region.

WHY THIS IS DIFFERENT

Three views of cost from three angles, recorded in a report reviewers and auditors can rely on. It reuses Module 01, which is why it costs so little.

3 HOW WE WILL DO IT

Step	Duration	What we do	Output
Build	4 weeks	Assemble peer database; build the three checks.	Assurance report
Validate	4 weeks	Shadow mode against recent sanctions; calibrate thresholds.	Calibrated checks

4 BENEFITS

For the Department

- Evidence-based verification on every DPR.
- Vigilance-ready trail of checks and overrides.

For consultants, contractors & citizens

- Engineers see where their estimate sits against peers.
- Fewer post-sanction revisions.

5 PRICE AND KEY RISKS

One-time price: ₹5–6 lakh (includes its share of the shared platform). Hosting, AI usage and maintenance are priced once for the whole platform.

Too many alerts early on	Shadow mode and calibration before go-live.
Too few comparable projects	The report says so, rather than making a weak comparison.

Digital Approval Tracking

DPR workflow with deadlines

Every DPR tracked from submission to sanction, with automatic escalation.

OWNER & PARTNERS RCD: all divisions and circles, CDO, Engineer-in-Chief, Secretariat; Finance	STATUS TODAY File-based. DPR files move Division → Circle → CDO → Engineer-in-Chief → Department → Finance. There is no DPR-specific tracker with stage deadlines (to be confirmed with the Department).
WHEN Phase 1 (months 1–4)	ONE-TIME PRICE ₹12–16 lakh · running costs shared across the platform

1 THE PROBLEM

- No one can see, at a glance, where every DPR is and how long it has been there.
- No stage deadlines are enforced; files can wait for weeks without escalation.
- Returned DPRs carry handwritten remarks that are hard to act on.

2 WHAT WE WILL BUILD

A DPR-specific workflow with stage deadlines, role-based queues and automatic escalation. It complements, and can link to, the state's e-Office.

- **Role-based queues:** Each officer sees only what needs action, with countdown timers.
- **Stage deadlines:** Agreed days per stage, with alerts at 80% and 100% of the time allowed.
- **Gatekeeping:** A DPR cannot enter the chain without its compliance and cost-assurance reports.
- **Clear returns:** Specific sections marked for correction, not generic remarks.

WHY THIS IS DIFFERENT

Not e-Office with a DPR label: stages cannot be skipped, deadlines escalate automatically, and approvals carry the evidence from Modules 03 and 04.

3 HOW WE WILL DO IT

Step	Duration	What we do	Output
Map	3 weeks	Document the real approval chain in 2 circles; set deadlines from observed data.	Process map
Pilot	8 weeks	Workflow and dashboards live in Ranchi circle; SMS alerts.	Live tracking
Roll out	6 weeks	All circles; Secretary's dashboard; training for all roles.	State-wide tracking

4 BENEFITS

For the Department

- Real-time pipeline for the Secretary and Engineer-in-Chief.
- Approval time targeted to fall from months to weeks.
- Bottleneck analytics by stage and office.

For consultants, contractors & citizens

- Submitters see status without phone calls.
- Faster correction and resubmission.

5 PRICE AND KEY RISKS

One-time price: ₹12–16 lakh (includes its share of the shared platform). Hosting, AI usage and maintenance are priced once for the whole platform.

Parallel paper files continue	Final sanction requires the digital record, so upstream adoption follows.
Weak connectivity in field offices	Light web app, SMS alerts, offline drafts that sync later.

Project Knowledge Base

Searchable archive of past projects

Twenty-five years of project memory, searchable in seconds.

OWNER & PARTNERS RCD: CDO, divisions, record rooms at Project Bhawan and circle offices	STATUS TODAY Paper archives. Past DPRs and sanctions since the state's formation in 2000 sit in record rooms. They cannot be searched or compared.
WHEN Phase 1 (months 1–4)	ONE-TIME PRICE ₹13–19 lakh · running costs shared across the platform

1 THE PROBLEM

- Knowledge leaves with officers on transfer; similar past work is hard to find.
- Engineers cannot quickly pull comparable DPRs for reference or cost comparison.
- Modules 01 and 04 need historical data that is not yet digital.

2 WHAT WE WILL BUILD

A digitised, AI-searchable archive of DPRs and sanction orders that becomes the data foundation for cost benchmarking.

- **Meaning-based search:** "Two-lane road with a minor bridge in hilly forest terrain" finds relevant projects even with different wording.
- **Comparable projects:** Per-km cost of the five most similar sanctioned projects for any new proposal.
- **Reference library:** Full past DPRs one click away while drafting.

WHY THIS IS DIFFERENT

Not a filing system: it understands what each project is, which is what makes similarity search and cost comparison possible.

3 HOW WE WILL DO IT

Step	Duration	What we do	Output
Inventory	3 weeks	Survey record rooms; prioritise recent, high-value and common project types.	Digitisation plan
Digitise	10 weeks	Scan and index up to ~800 DPRs; build search; pilot with engineers.	Searchable archive
Run	Ongoing	Every newly sanctioned DPR added automatically.	Self-growing archive

4 BENEFITS

For the Department

- Institutional memory survives transfers.
- Feeds Modules 01 and 04, improving accuracy over time.

For consultants, contractors & citizens

- Consultants calibrate estimates against approved precedent.
- Fewer disputes about past approvals.

5 PRICE AND KEY RISKS

One-time price: ₹13–19 lakh (build incl. its share of the shared platform ₹9–12 lakh; digitisation of up to ~800 past DPRs ₹4–7 lakh). Hosting, AI usage and maintenance are priced once for the whole platform.

Poor condition of old files	Start with digital and recent files; manual correction where OCR fails.
Circles reluctant to share	Read-only central access; circles keep ownership.

GIS Link-up

DPR pipeline on RCD's existing map

Every proposal and sanction placed on the road map RCD already uses.

OWNER & PARTNERS RCD GIS cell; CDO; Forest, Environment & Climate Change Department (layers); JAP-IT	STATUS TODAY GIS portal exists. RCD already runs a GIS portal for State Highways and Major District Roads. It is not linked to DPRs in preparation or under approval.
WHEN Phase 2 (months 5–10)	ONE-TIME PRICE ₹6–8 lakh · running costs shared across the platform

1 THE PROBLEM

- Planning decisions use lists and tables; proposals are not visible on the network map.
- Two proposals on the same stretch, or proposals crossing forest land and elephant corridors, are spotted late.

2 WHAT WE WILL BUILD

We extend, not replace, the existing RCD GIS: DPR and approval-stage layers plus early conflict checks.

- **Pipeline layer:** Proposed, under-approval and sanctioned DPRs on their road segments, with cost and stage.
- **Overlap alerts:** Automatic flag when two proposals target the same stretch.
- **Sensitivity overlay:** Forest land and wildlife-corridor layers (where shared) to flag clearance needs at DPR stage.

WHY THIS IS DIFFERENT

Reuses what the state has already paid for. The value is in the link to the live DPR pipeline, not in another map.

3 HOW WE WILL DO IT

Step	Duration	What we do	Output
Integrate	4 weeks	Connect to RCD GIS; locate existing DPRs on segments.	Linked layers
Extend	4 weeks	Overlap alerts, sensitivity overlay, district and circle views.	Planning map

4 BENEFITS

For the Department

- Duplicate spending on the same stretch avoided.
- Forest and corridor conflicts caught before DPRs are finalised.

For consultants, contractors & citizens

- Contractors see upcoming work by district.
- Citizens can see which roads are being improved.

5 PRICE AND KEY RISKS

One-time price: ₹6–8 lakh (includes its share of the shared platform). Hosting, AI usage and maintenance are priced once for the whole platform.

Limited access to the GIS portal	Integration agreed with the RCD GIS cell and JAP-IT in week one; fallback is a lightweight layer on open base maps.
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How we will work with the Department

Start small, prove value	A 6–8 week proof of concept on RCD's own data in one division, adjusted against Phase 1. Scale only after results are visible.
Transparent procurement	Through GeM, JAP-IT or an empanelled system integrator; open tender for the full rollout where rules require it.
Data stays with the state	Hosting in the Jharkhand State Data Centre (JAP-IT) or a MeitY-empanelled Indian cloud; compliant with the DPDP Act 2023.
Security first	A CERT-In-empanelled audit before go-live; role-based access; tamper-evident audit logs.
No vendor lock-in	Open-source technology, source code in escrow, full documentation and training for RCD engineers.
Hindi and English	All screens, reports and training material in Hindi and English.
Pay for outcomes	Milestone payments tied to measurable results: DPRs processed, approval days, benchmark accuracy.
Built for Jharkhand	On-site presence in Ranchi during rollout; internships for students from Jharkhand engineering colleges.

ONE PLATFORM, MORE DEPARTMENTS

The same platform can later serve the Rural Works Department (₹5,082 crore in 2026–27) and the Building Construction Department with configuration rather than a new build, so each extension costs a fraction of the first.

i SOURCES (PUBLIC INFORMATION, ACCESSED SEPTEMBER 2026)

1. RCD Jharkhand: road network achievements (12,736 km; SH 1,231.9 km; MDR 4,845.7 km) — <https://rcd.jharkhand.gov.in/achievements.php>
2. RCD Jharkhand: organisation (Engineer-in-Chief; CE, Central Design Organisation) — <https://rcd.jharkhand.gov.in/organisation.php>
3. RCD GIS portal — https://gis.jharkhand.gov.in/rcd_portal/index.php
4. Jharkhand Budget 2026–27 (Drishti IAS): RCD ₹6,601.28 crore; RWD ₹5,081.74 crore — <https://www.drishtias.com/state-pcs-current-affairs/jharkhand-budget-202627>
5. Jharkhand Budget Analysis 2026–27 (PRS India) — <https://prsindia.org/budgets/states/jharkhand-budget-analysis-2026-27>
6. Jharkhand Budget 2026–27 overview (Elets eGov) — <https://egov.eletsonline.com/2026/02/jharkhand-budget-2026-27-unveils-%E2%82%B91-58-lakh-crore-plan-focused-on-welfare-infrastructure-and-economic-growth/>
7. Jharkhand RCD Schedule of Rates status (CivilShape) — <https://www.civilshape.in/2025/06/Jharkhand-Road-Construction-Department-SOR-Updated-in-2025-Jharkhand-RCD-SOR.html>
8. State-wise Schedule of Rates 2026 (InfraLens) — <https://infralens.in/knowledge/state-wise-sor-india-2026>
9. Jharkhand State Roads Project: forest land along project roads (ADB) — <https://www.adb.org/sites/default/files/project-documents/40005-ind-siee.pdf>
10. JAP-IT: nodal e-governance agency, State Data Centre — <https://japit.jharkhand.gov.in/>

Status descriptions are based on public information up to September 2026 and should be confirmed with the Department before formal submission.